

Restricciones de $\geq$

$$\text{MAX: } Z = 4 x_1 + 3 x_2$$

$$\left\{ \begin{array}{l} 6 x_1 + 16 x_2 \leq 48.000 \\ 12 x_1 + 6 x_2 \leq 42.000 \\ x_2 \geq 1.500 \end{array} \right.$$

$$x_1, x_2 \geq 0$$

Forma estándar

$$\text{MAX: } Z = 4 x_1 + 3 x_2$$

$$\left\{ \begin{array}{rcl} 6 x_1 + 16 x_2 + x_3 & = & 48.000 \\ 12 x_1 + 6 x_2 + x_4 & = & 42.000 \\ x_2 - x_5 & = & 1.500 \end{array} \right.$$

$$x_1, x_2, x_3, x_4, x_5 \geq 0$$

PRIMERA BASE DEL “SIMPLEX”

$$\left\{ \begin{array}{rcl} \mathbf{x_3} & = & 48.000 \\ \mathbf{x_4} & = & 42.000 \\ -\mathbf{x_5} & = & 1.500 \end{array} \right. \Rightarrow \text{VULNERA EL PRINCIPIO DE NN}$$

$$\mathbf{x_1, x_2, x_3, x_4, x_5 \geq 0}$$

PRIMERA BASE DEL “SIMPLEX”

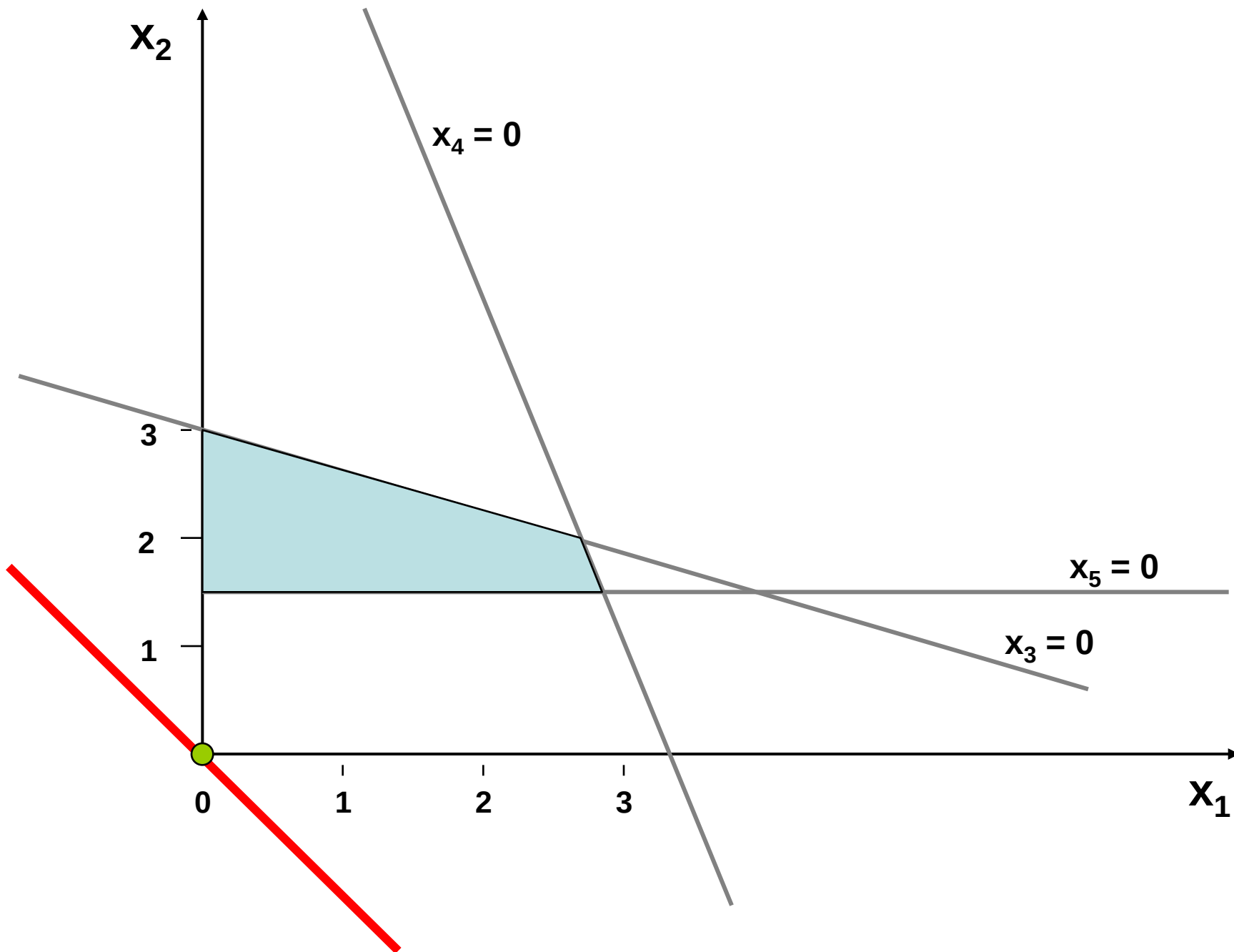
$$\text{MAX: } Z = 4 x_1 + 3 x_2 - M \cdot \mu_1$$

$$\left\{ \begin{array}{rcl} x_3 & & = 48.000 \\ & x_4 & = 42.000 \\ & -x_5 + \mu_1 & = 1.500 \end{array} \right.$$

$$x_1, x_2, x_3, x_4, x_5, \mu_1 \geq 0$$

PROBLEMA DE MINIMIZACIÓN

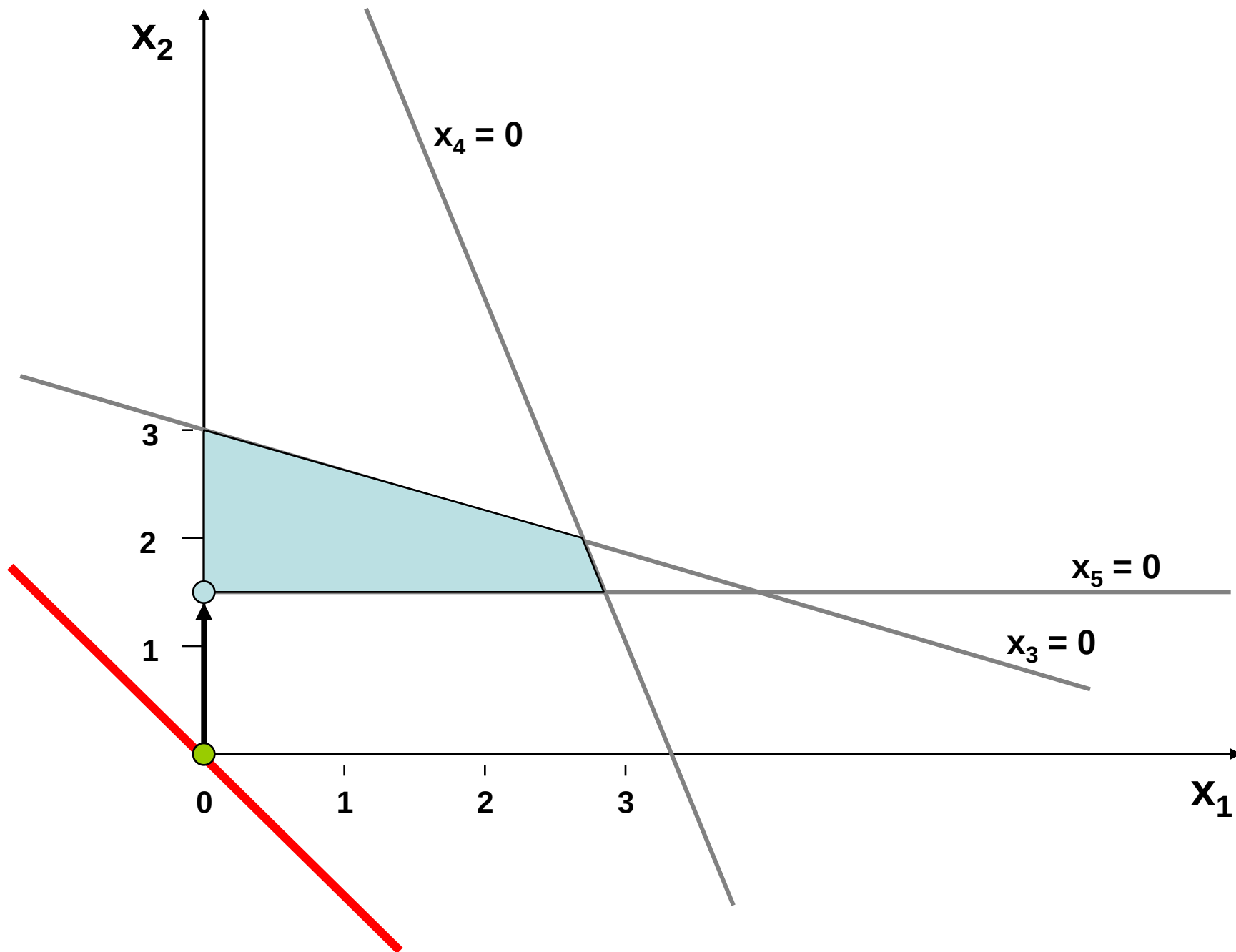
$$\text{MIN: } Z = \dots + M \cdot \mu_1 \dots$$



		c_j	4	3	0	0	0	-M	
c_k	x_k	B	A_1	A_2	A_3	A_4	A_5	A_μ	b_i/a_{ij}
0	x_3	48.000	6	16	1				3.000
0	x_4	42.000	12	6		1			7.000
-M	μ	1.500		1			-1	1	1.500
$Z = -1.500 M$			-4	-M-3	0	0	M	0	

↑

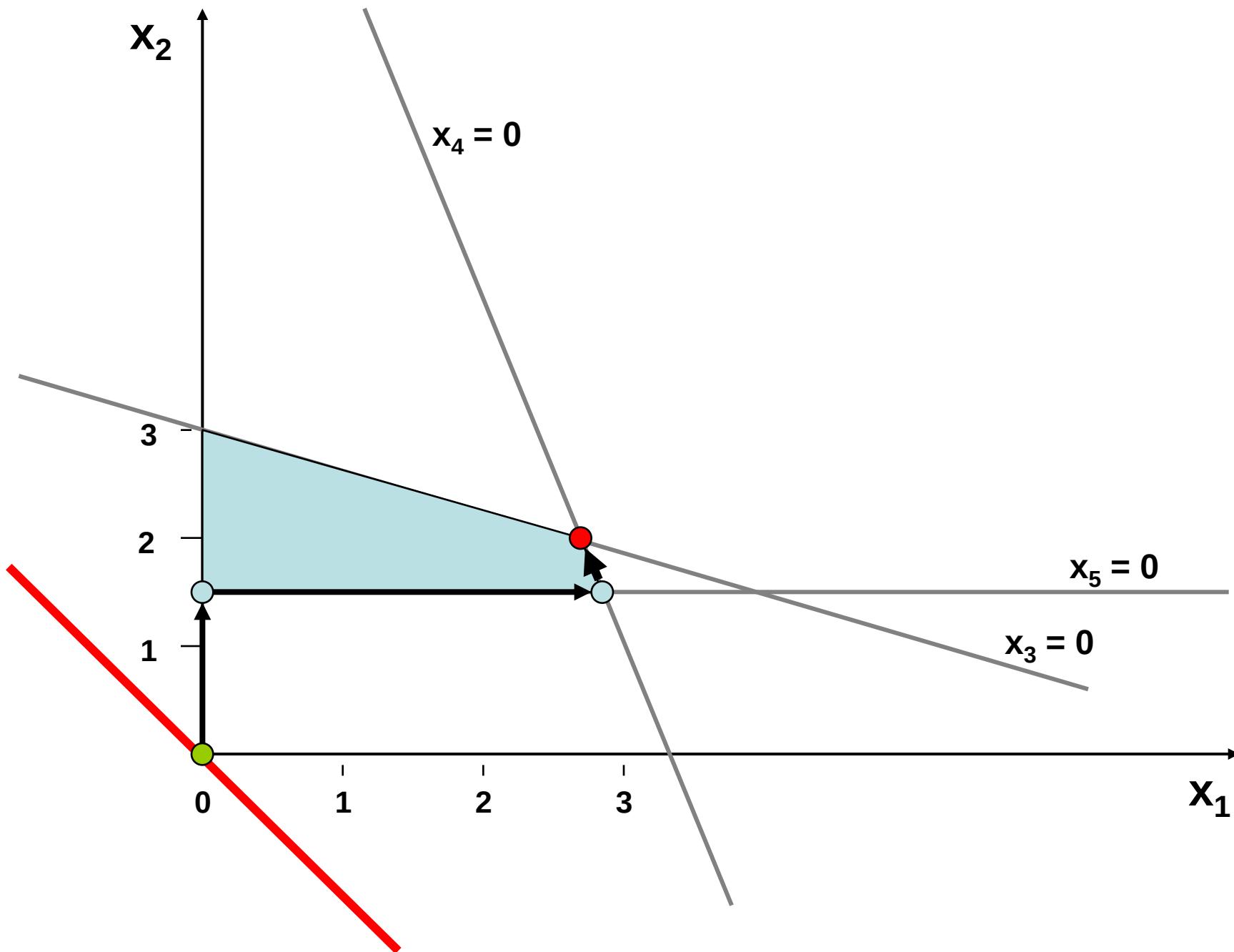
		c_j	4	3	0	0	0	-M	
c_k	x_k	B	A_1	A_2	A_3	A_4	A_5	A_μ	b_i/a_{ij}
0	x_3	24.000	6		1		16	-16	4.000
0	x_4	33.000	12			1	6	-6	2.750
3	x_2	1.500		1			-1	1	---
$Z = 4.500$			-4	0	0	0	-3	3+M	



		c_j	4	3	0	0	0	-M	
c_k	x_k	B	A_1	A_2	A_3	A_4	A_5	A_μ	b_i/a_{ij}
0	x_3	24.000	6		1		16	-16	4.000
0	x_4	33.000	12			1	6	-6	2.750
3	x_2	1.500		1			-1	1	---
$Z = 4.500$			-4	0	0	0	-3	$3+M$	

0	x_3	576,92			1	-0,5	13	-13	576,9231
4	x_1	5.500	1			0,0833	0,5	-0,5	5.500
3	x_2	1.500		1		0	-1	1	---
$Z = 15.500$			0	0	0	0,3333	-1	$1+M$	

0	x_5	576,9231			0,0769	-0,0385	1	-1	
4	x_1	2.461,539	1		-0,0385	0,1026			
3	x_2	2076,923		1	0,0769	-0,0385			
$Z = 16.076,92$			0	0	0,0769	0,2949	0	M	



Restricciones de =

MAX: $Z = 4 x_1 + 3 x_2$

$$\left\{ \begin{array}{l} 6 x_1 + 16 x_2 \leq 48.000 \\ 12 x_1 + 6 x_2 \leq 42.000 \\ x_2 = 1.500 \end{array} \right.$$

$$x_1, x_2 \geq 0$$

Forma estándar

$$\text{MAX: } Z = 4 x_1 + 3 x_2$$

$$\left\{ \begin{array}{lcl} 6 x_1 + 16 x_2 + x_3 & = & 48.000 \\ 12 x_1 + 6 x_2 + x_4 & = & 42.000 \\ x_2 & = & 1.500 \end{array} \right.$$

$$x_1, x_2, x_3, x_4 \geq 0$$

PRIMERA BASE DEL “SIMPLEX”

$$\left\{ \begin{array}{rcl} \mathbf{x}_3 & = & 48.000 \\ \mathbf{x}_4 & = & 42.000 \\ 0 & = & 1.500 \end{array} \right. \Rightarrow \text{NO ES COMPATIBLE}$$

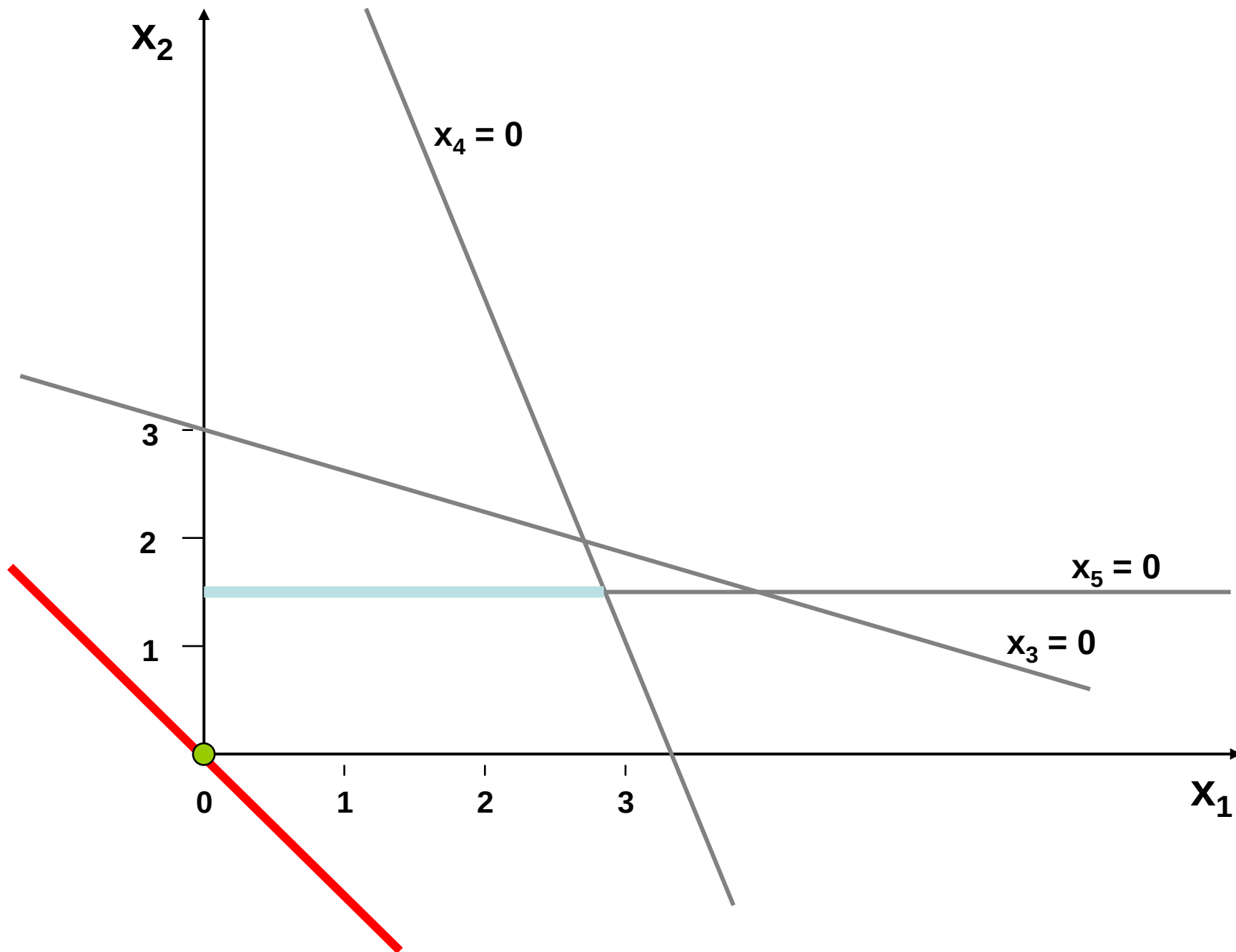
$$\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, \mathbf{x}_4 \geq 0$$

PRIMERA BASE DEL “SIMPLEX”

$$\text{MAX: } Z = 4 x_1 + 3 x_2 - M \lambda_1$$

$$\left\{ \begin{array}{rcl} x_3 & & = 48.000 \\ & x_4 & = 42.000 \\ & & + \lambda_1 = 1.500 \end{array} \right.$$

$$x_1, x_2, x_3, x_4, \lambda_1 \geq 0$$



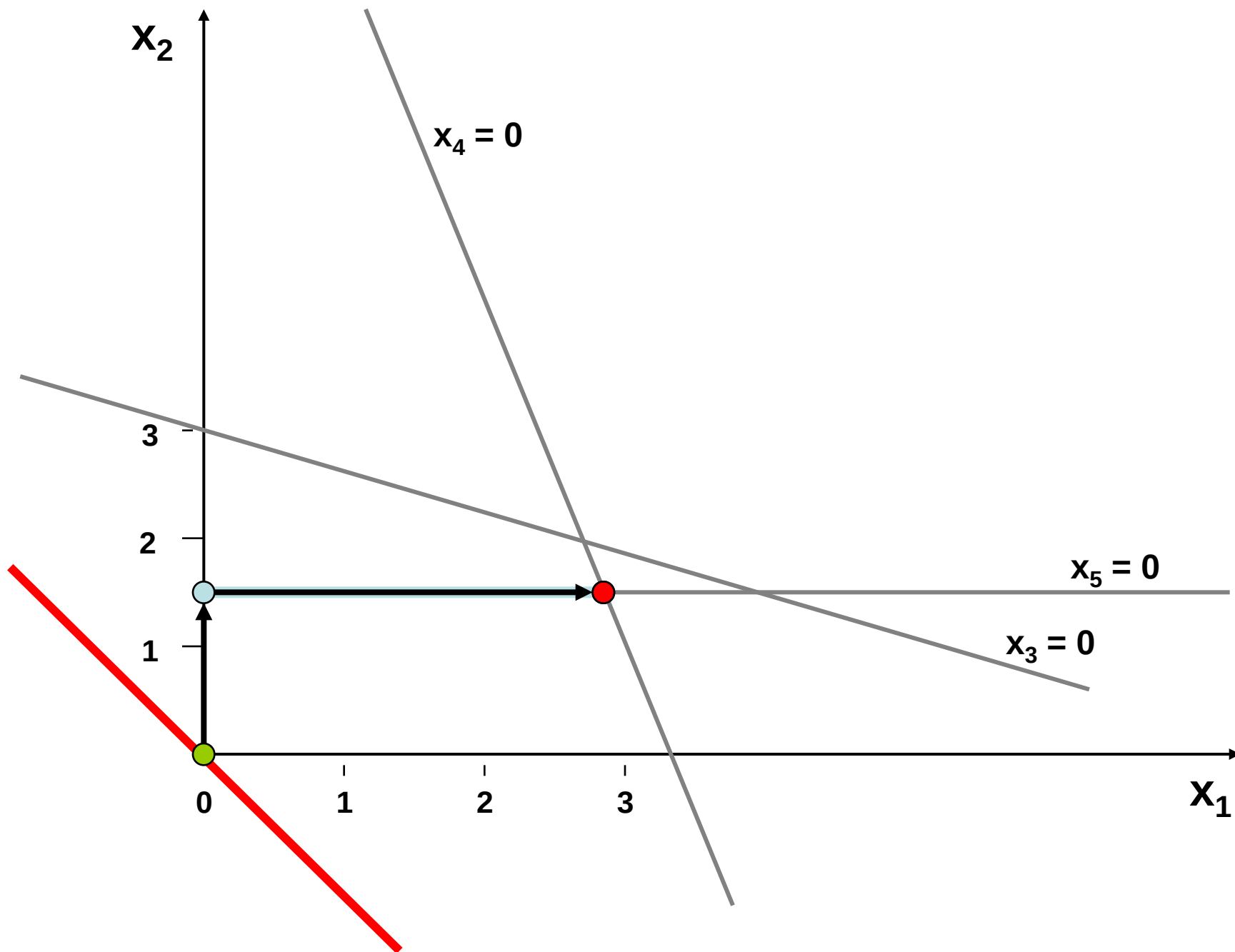
		c_j	4	3	0	0	-M	
c_k	x_k	B	A_1	A_2	A_3	A_4	A_λ	b_i/a_{ij}
0	x_3	48.000	6	16	1			3.000
0	x_4	42.000	12	6		1		7.000
 -M	λ	1.500		1			1	1.500
Z = - M 1.500			-4	-M-3	0	0	3+M	



0	x_3	24.000	6		1		16	4.000
 0	x_4	33.000	12			1	6	2.750
3	x_2	1.500		1			-1	---
Z = 4.500			-4	0	0	0	-3+M	



0	x_3	7.500			1	-0,5	-13	
4	x_1	2.750	1			0,0833	-0,5	
3	x_2	1.500		1		0	1	
Z = 15.500			0	0	0	0,3333	1+M	



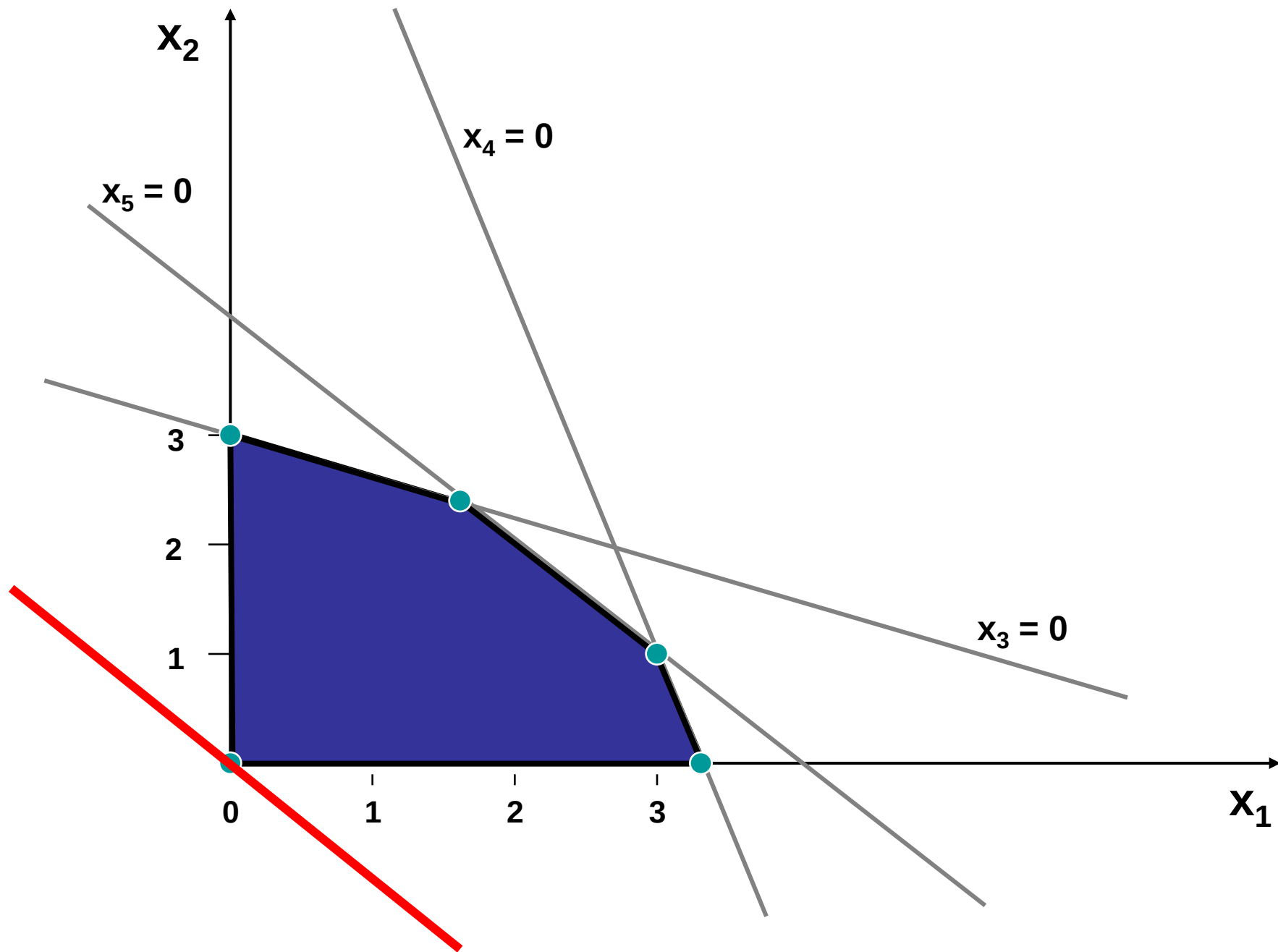
CASOS PARTICULARES

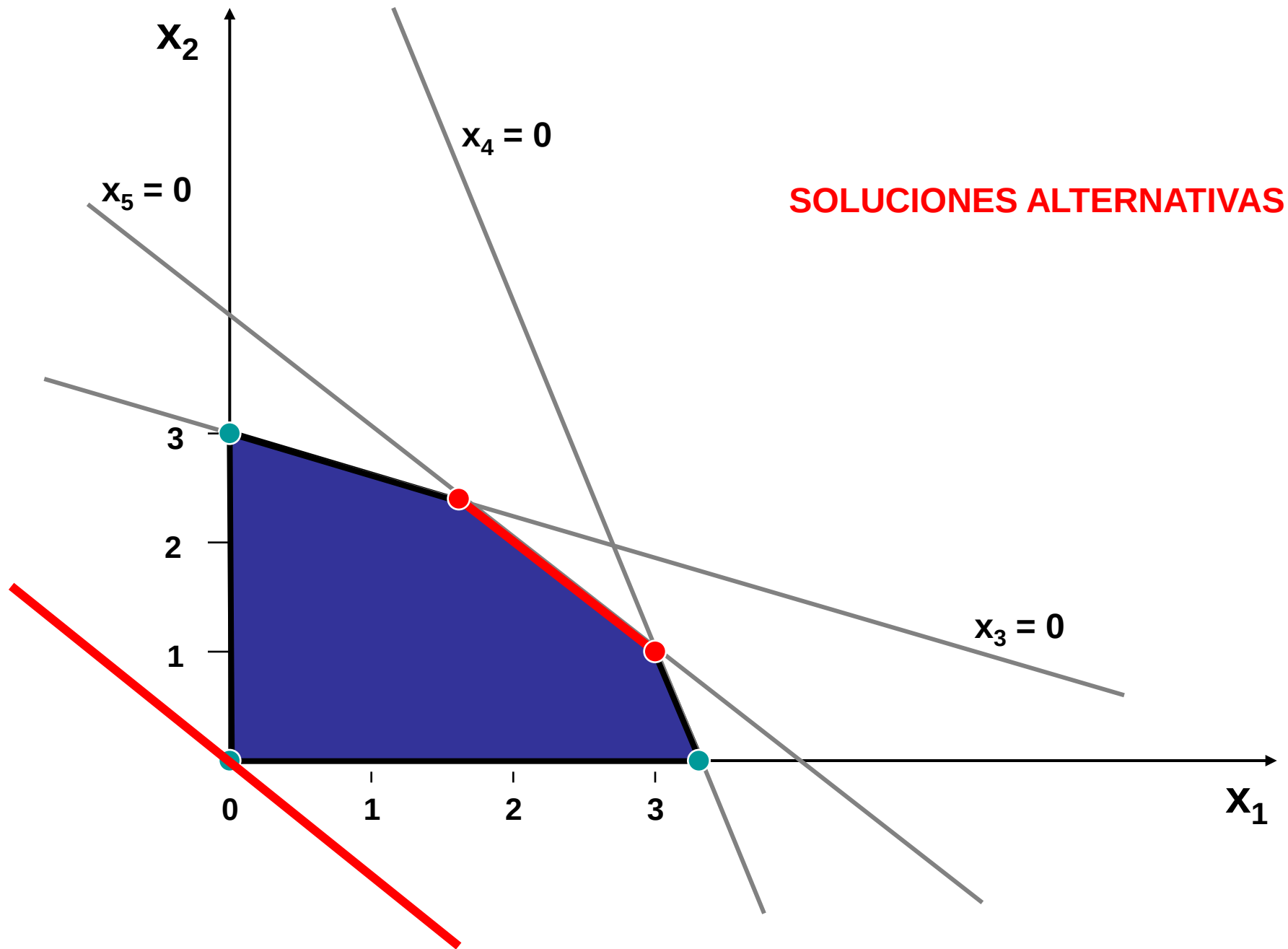
- SOLUCIÓN ALTERNATIVA
- SOLUCIÓN DEGENERADA
- POLITOPO ABIERTO
- SOLUCIÓN INCOMPATIBLE

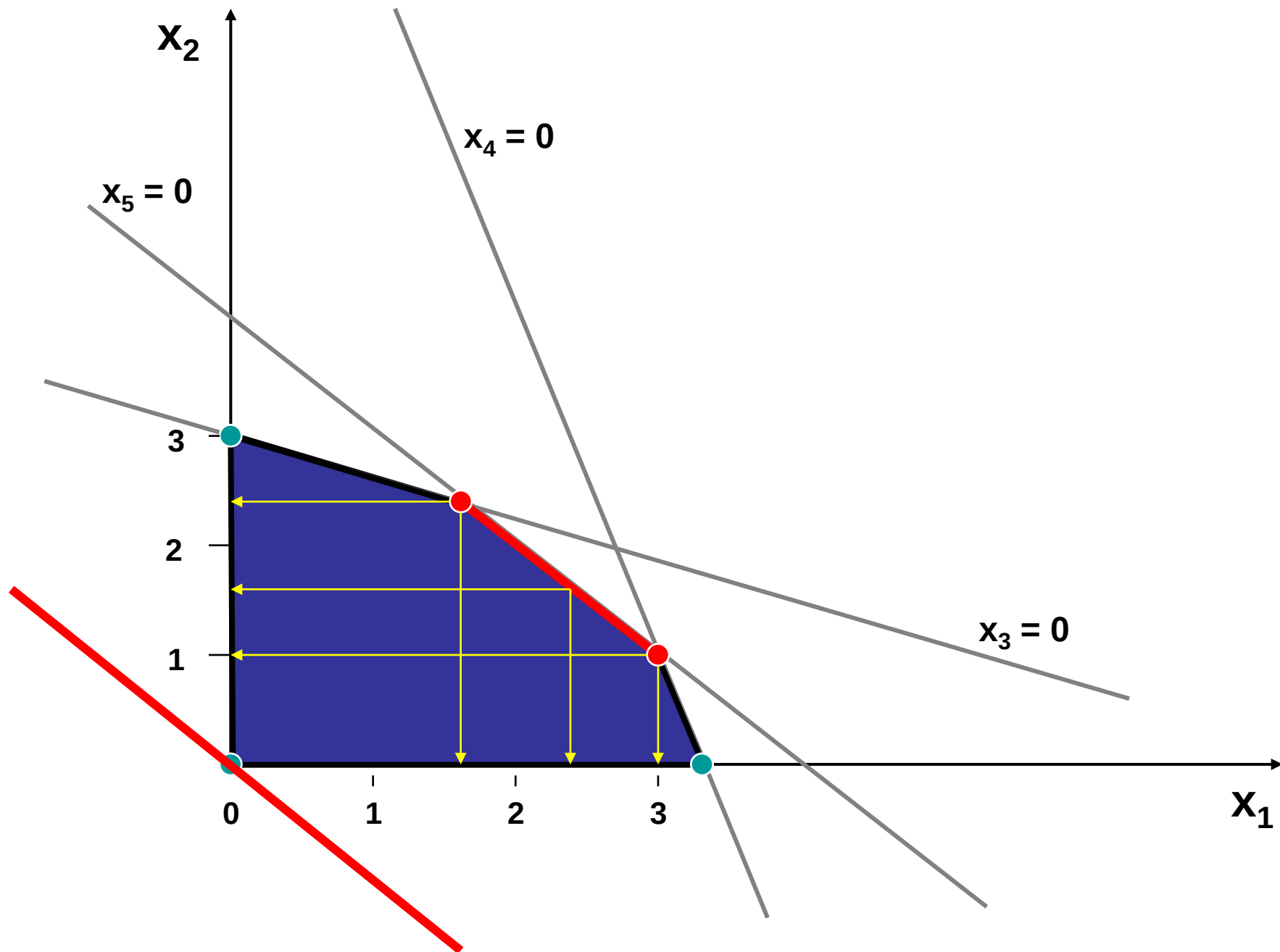
$$\text{MAX: } Z = 3 x_1 + 3 x_2$$

$$\left\{ \begin{array}{l} 6 x_1 + 16 x_2 \leq 48.000 \\ 12 x_1 + 6 x_2 \leq 42.000 \\ 9 x_1 + 9 x_2 \leq 36.000 \end{array} \right.$$

$$x_1, x_2 \geq 0$$







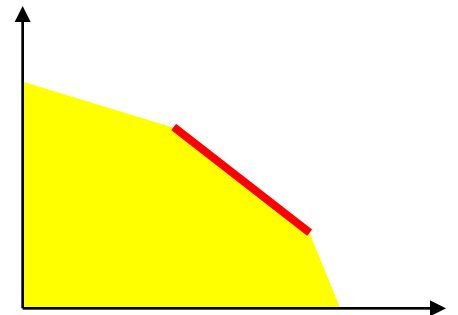
		c_j	3	3	0	0	0		
	c_k	x_k	B	A_1	A_2	A_3	A_4	A_5	b_i/a_{ij}
←	0	x_3	14.000			1	5/3	-26/9	8.400
	3	x_1	3.000	1			1/6	-1/9	18.000
	3	x_2	1.000		1		-1/6	2/9	---
	Z = 12.000			0	0	0	0*	3/9	

		c_j	3	3	0	0	0	
c_k	x_k	B	A₁	A₂	A₃	A₄	A₅	b_i/a_{ij}
0	x_3	14.000			1	5/3	-26/9	8.400
3	x_1	3.000	1			1/6	-1/9	18.000
3	x_2	1.000		1		-1/6	2/9	---
Z = 12.000			0	0	0	0*	3/9	

0	x_4	8.400			3/5	1	-26/15	
3	x_1	1.600	1		-1/10		16/90	
3	x_2	2.400		1	1/10		-1/15	
Z = 12.000			0	0	0*	0	1/3	

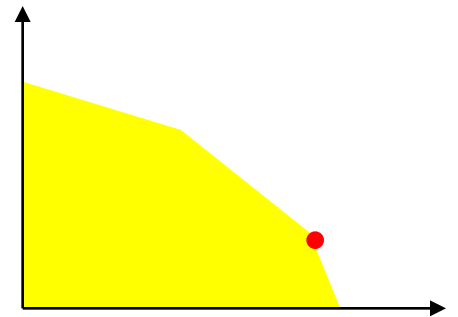
$$X = \alpha \cdot \begin{pmatrix} 3.000 \\ 1.000 \\ 14.000 \\ 0 \\ 0 \end{pmatrix} + (1 - \alpha) \cdot \begin{pmatrix} 1.600 \\ 2.400 \\ 0 \\ 8.400 \\ 0 \end{pmatrix}$$

$$0 \leq \alpha \leq 1$$



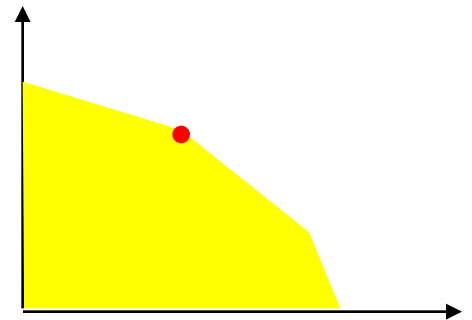
Ejemplo: $\alpha = 1$

$$X = 1 \cdot \begin{pmatrix} 3.000 \\ 1.000 \\ 14.000 \\ 0 \\ 0 \end{pmatrix} + 0 \cdot \begin{pmatrix} 1.600 \\ 2.400 \\ 0 \\ 8.400 \\ 0 \end{pmatrix} = \begin{pmatrix} 3.000 \\ 1.000 \\ 14.000 \\ 0 \\ 0 \end{pmatrix}$$



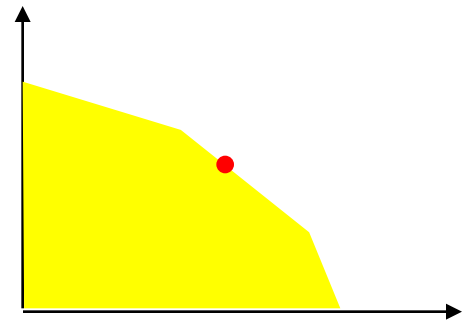
Ejemplo: $\alpha = 0$

$$X = 0 \cdot \begin{pmatrix} 3.000 \\ 1.000 \\ 14.000 \\ 0 \\ 0 \end{pmatrix} + 1 \cdot \begin{pmatrix} 1.600 \\ 2.400 \\ 0 \\ 8.400 \\ 0 \end{pmatrix} = \begin{pmatrix} 1.600 \\ 2.400 \\ 0 \\ 8.400 \\ 0 \end{pmatrix}$$



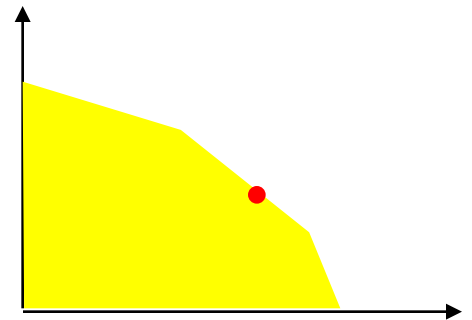
Ejemplo: $\alpha = 0,3$

$$X = 0,3 \cdot \begin{pmatrix} 3.000 \\ 1.000 \\ 14.000 \\ 0 \\ 0 \end{pmatrix} + 0,7 \cdot \begin{pmatrix} 1.600 \\ 2.400 \\ 0 \\ 8.400 \\ 0 \end{pmatrix} = \begin{pmatrix} 2.020 \\ 1.980 \\ 4.200 \\ 5.880 \\ 0 \end{pmatrix}$$



Ejemplo: $\alpha = 0,5$

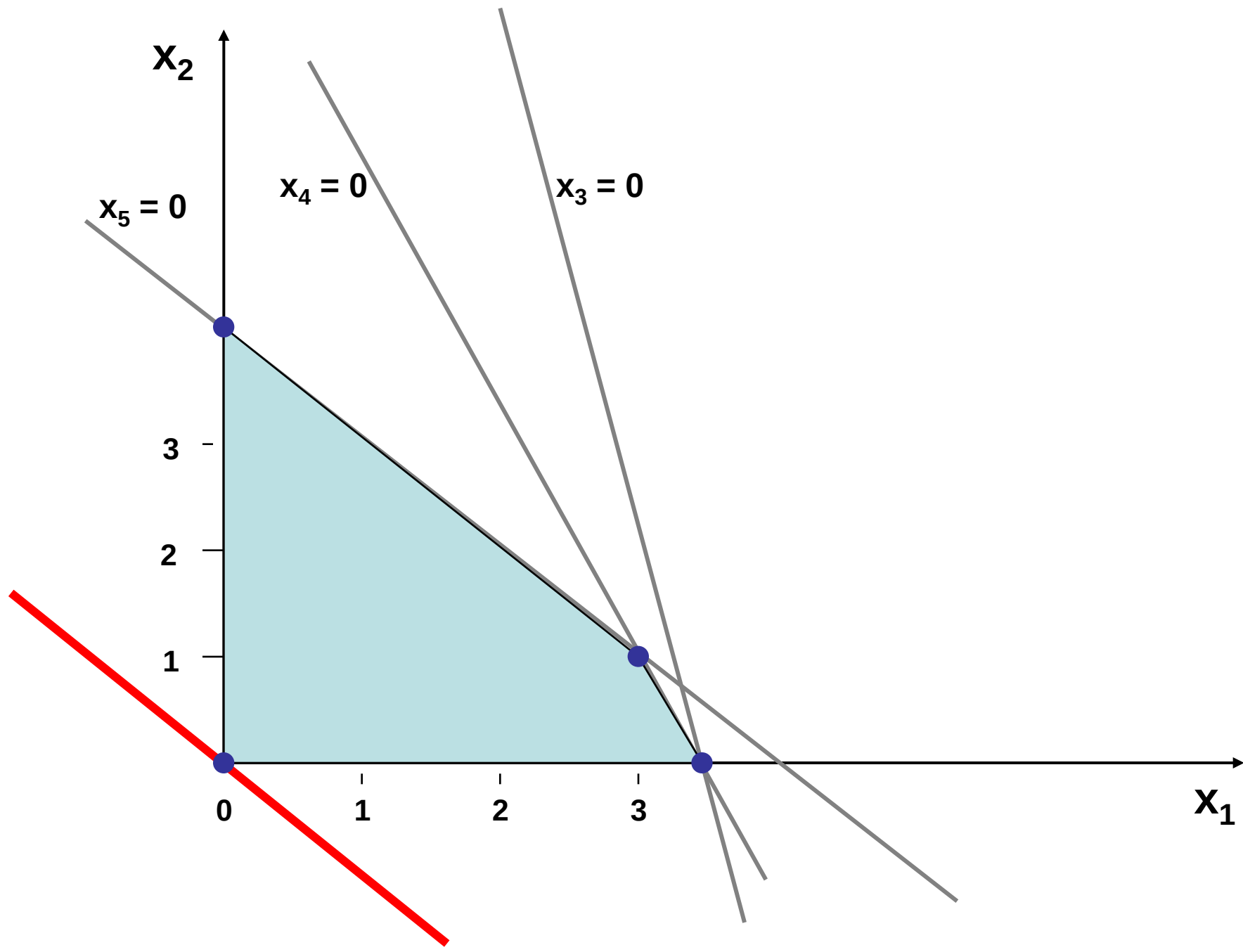
$$X = 0,5 \cdot \begin{pmatrix} 3.000 \\ 1.000 \\ 14.000 \\ 0 \\ 0 \end{pmatrix} + 0,5 \cdot \begin{pmatrix} 1.600 \\ 2.400 \\ 0 \\ 8.400 \\ 0 \end{pmatrix} = \begin{pmatrix} 2.300 \\ 1.700 \\ 7.000 \\ 4.200 \\ 0 \end{pmatrix}$$



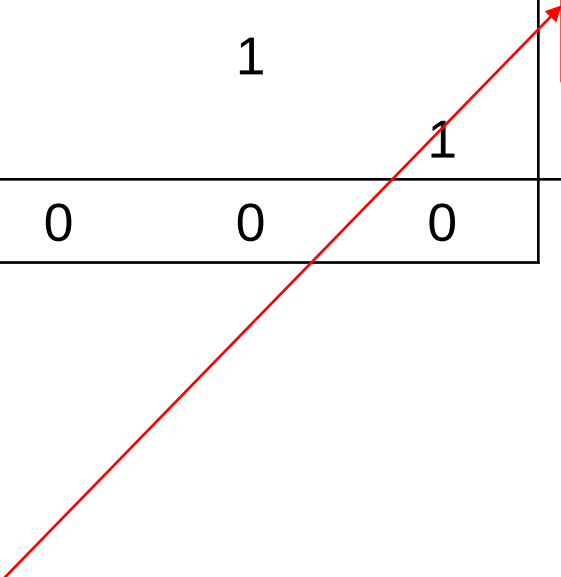
$$\text{MAX: } Z = 4 x_1 + 3 x_2$$

$$\left\{ \begin{array}{l} 10 x_1 + 4 x_2 \leq 35.000 \\ 12 x_1 + 6 x_2 \leq 48.000 \\ 9 x_1 + 9 x_2 \leq 36.000 \end{array} \right.$$

$$x_1, x_2 \geq 0$$



		c_j	4	3	0	0	0	
c_k	x_k	B	A₁	A₂	A₃	A₄	A₅	b_i/a_{ij}
0	x_3	35.000	10	4	1			3.500
0	x_4	42.000	12	6		1		3.500
0	x_5	36.000	9	9			1	4.000
$Z = 0$			-4	-3	0	0	0	




EMPATE DE Θ

		c_j	4	3	0	0	0	
c_k	x_k	B	A_1	A_2	A_3	A_4	A_5	b_i/a_{ij}
4	x_1	3.500	1	0,4	0,1			8.750
0	x_4	0		1,2	-1,2	1		0
0	x_5	4.500		5,4	-0,9		1	833,33
$Z = 14.000$			0	-1,4	0,4	0	0	

4	x_1	3.500	1		0,5	-0,333		7.000
3	x_2	0		1	-1	0,8333		0
0	x_5	4.500			4,5	-4,5	1	1.000
$Z = 14.000$			0	0	-1	1,1667	0	

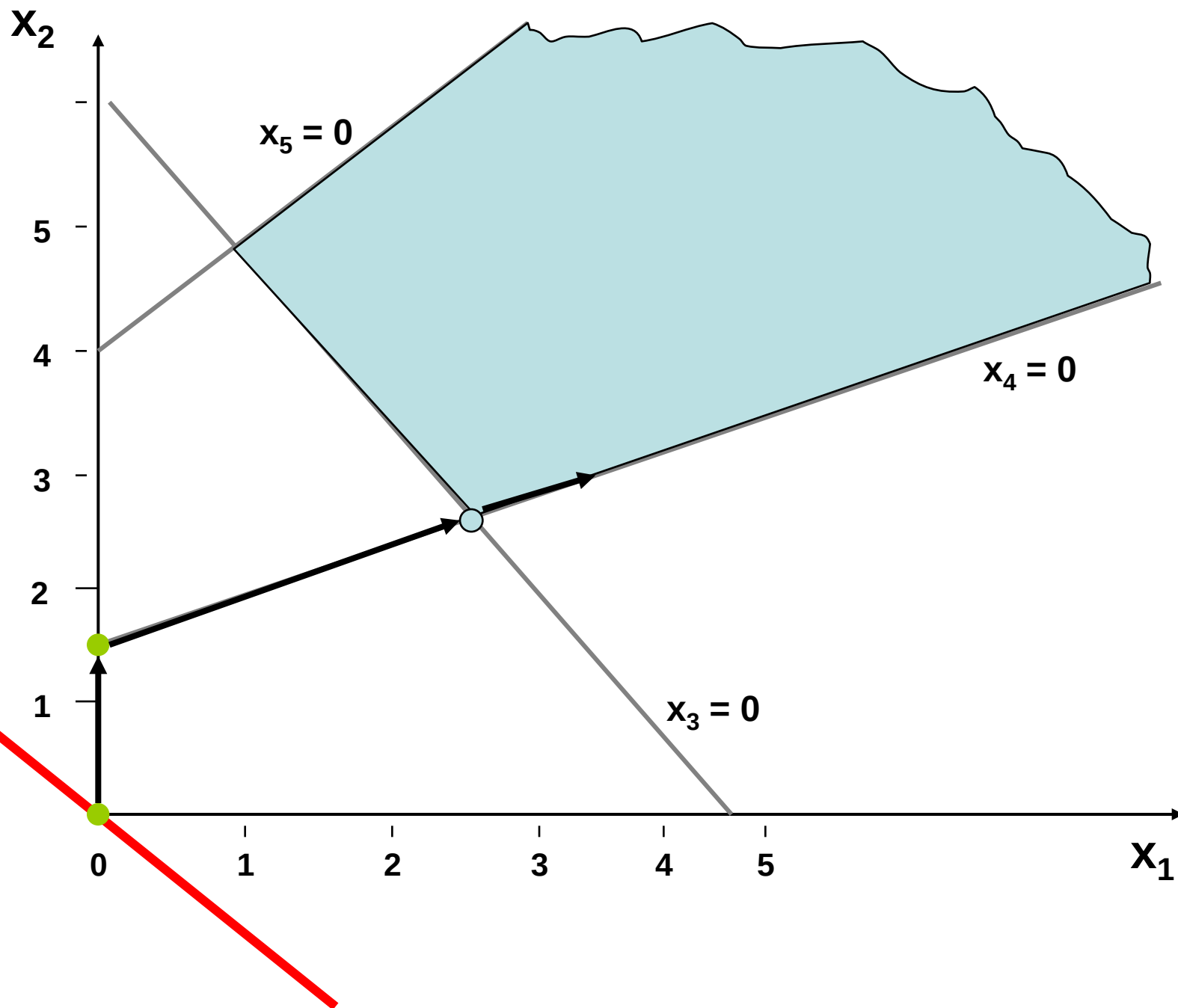
4	x_1	3000	1			0,1667	-0,111	
3	x_2	1.000		1		-0,1667	0,2222	
0	x_3	1.000			1	-1	0,2222	
$Z = 15.000$			0	0	0	0,1667	0,2222	

POLITOPO ABIERTO

$$\text{MAX: } Z = 4 x_1 + 3 x_2$$

$$\left\{ \begin{array}{l} 10 x_1 + 8 x_2 \geq 48.000 \\ -4 x_1 + 10 x_2 \geq 16.000 \\ -9 x_1 + 9 x_2 \leq 36.000 \end{array} \right.$$

$$x_1, x_2 \geq 0$$



3a. TABLA

		c_j	4	3	0	0	0	
c_k	x_k	B	A_1	A_2	A_3	A_4	A_5	b_i/a_{ij}
4	x_1	2.666,67	1		-0,0758	0,0606		---
3	x_2	2.666,67		1	-0,0303	-0,0758		---
0	x_5	36.000			-0,4091	1,2327	1	---
$Z = 18.666,67$			0	0	-0,3939	0,0152	0	



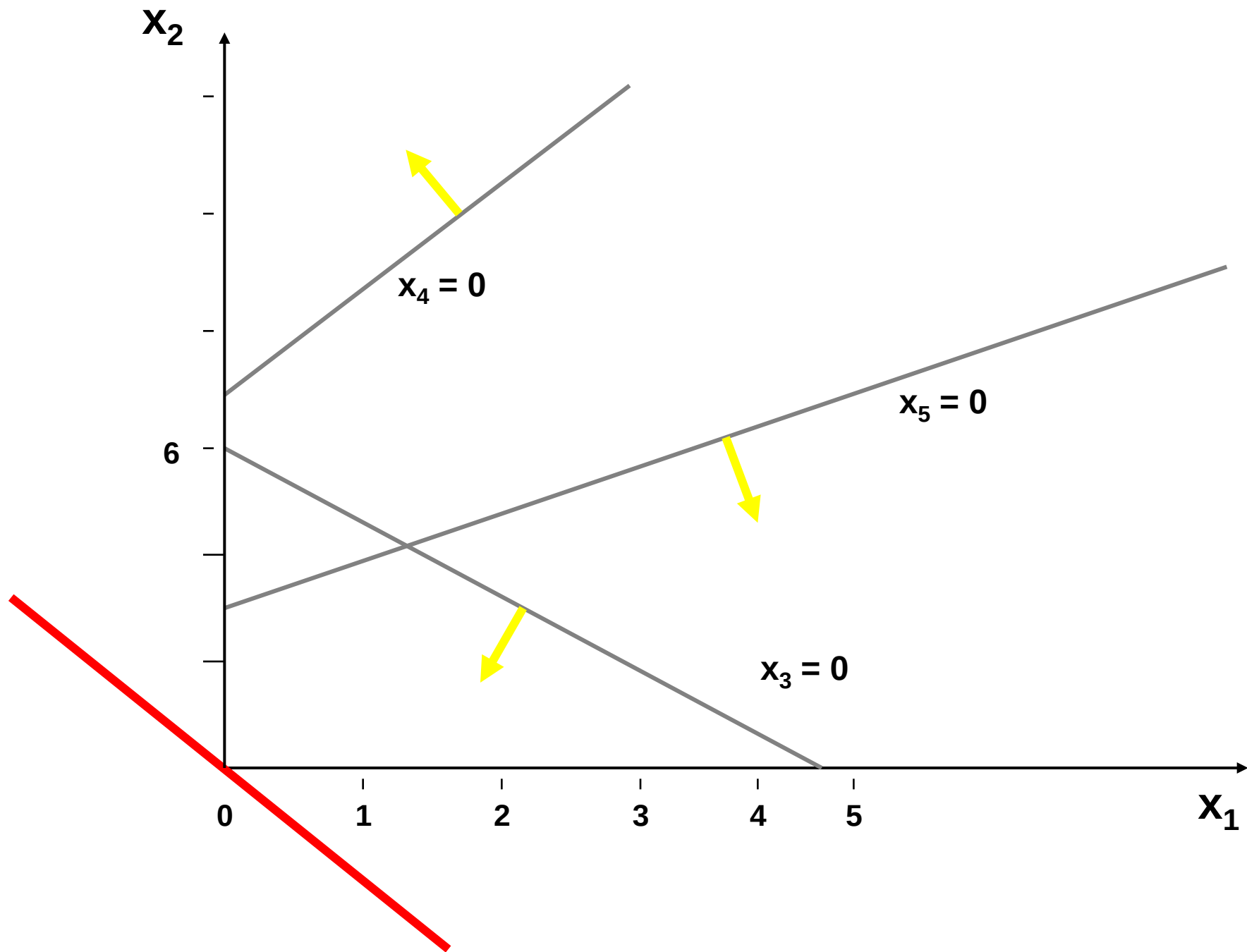
NO HAY NINGÚN θ

SOLUCIÓN INCOMPATIBLE

$$\text{MAX: } Z = 4 x_1 + 3 x_2$$

$$\left\{ \begin{array}{l} 10 x_1 + 8 x_2 \geq 48.000 \\ -12 x_1 + 6 x_2 \geq 42.000 \\ -9 x_1 + 9 x_2 \leq 36.000 \end{array} \right.$$

$$x_1, x_2 \geq 0$$



3a. TABLA

		c_j	4	3						- M	- M
c_k	x_k	B	A_1	A_2	A_3	A_4	A_5	$A_{\mu 1}$	$A_{\mu 2}$		
4	x_1	888,89	1		-0,0556		-0,0494	0,0556			
-M	μ_2	23.333,33			-0,3333	-1	-0,9630	0,3333	1		
3	x_2	4890		1	-0,0556		0,0617	0,0556			
$Z = 23.333,33M + 18.222,22$			0	0	$-0,3889 + 0,333 M$	M	$-0,0123 + 0,968 M$	$0,3889 + 0,6667 M$	0		

VARIABLE
ARTIFICIAL EN
LA BASE

TODOS LOS $z_j - c_j$ POSITIVOS